LAPORAN PRAKTIKUM JOBSHEET 2: DATABASE OPERASIONAL

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I. Tugas 1: Import Data dan Analisis Struktur Database

1. **Proses Import Data**
   - Langkah-langkah import data ke MySQL.
   - Verifikasi keberhasilan import data.
2. **Analisis Struktur Database**
   - Identifikasi tabel dan jumlah field:

Nama Tabel	Jumlah Field
productlines	3
products	5
orders	6
orderdetails	4
customers	7
employees	6
offices	5
payments	4

- Analisa hubungan antar tabel:

Tabel 1	Tabel 2	Jenis Relasi
productlines	poducts	One to Many
products	orderdetails	One to Many
orders	orderdetails	One to Many
customers	orders	One to Many
employees	customers	One to Many
offices	employees	One to Many
payments	customers	One to Many

II. Tugas 2: Analisis Hirarki Organisasi

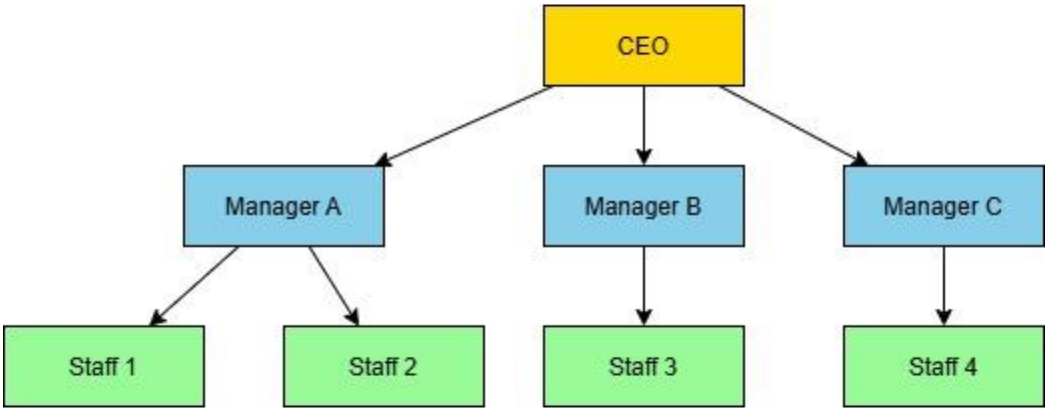
Eksekusi Query untuk Menampilkan Hirarki Pegawai

```
SELECT manager.employeeNumber AS id_manager,
       CONCAT(manager.firstName, " ", manager.lastName) AS Manager,
       employee.employeeNumber AS id_staff,
       CONCAT(employee.firstName, " ", employee.lastName) AS staff
FROM employees employee, employees manager
WHERE employee.reportsTo = manager.employeeNumber
ORDER BY manager.firstName;
```

id_manager	Manager	id_staff	Staff
1143	Anthony Bow	1165	Leslie Jennings
1143	Anthony Bow	1166	Leslie Thompson
1143	Anthony Bow	1188	Julie Firrelli
1143	Anthony Bow	1216	Steve Patterson
1143	Anthony Bow	1286	Foon Yue Tseng
1143	Anthony Bow	1323	George Vanauf
1002	Diane Murphy	1056	Mary Patterson
1002	Diane Murphy	1076	Jeff Firrelli
1102	Gerard Bondur	1337	Loui Bondur
1102	Gerard Bondur	1370	Gerard Hernandez
1102	Gerard Bondur	1401	Pamela Castillo
1102	Gerard Bondur	1501	Larry Bott
1102	Gerard Bondur	1504	Barry Jones
1102	Gerard Bondur	1702	Martin Gerard
1621	Mami Nishi	1625	Yoshimi Kato
1056	Mary Patterson	1088	William Patterson
1056	Mary Patterson	1102	Gerard Bondur
1056	Mary Patterson	1143	Anthony Bow
1056	Mary Patterson	1621	Mami Nishi
1088	William Patterson	1611	Andy Fixter
1088	William Patterson	1612	Peter Marsh
1088	William Patterson	1619	Tom King

Hasil dan Analisis Hirarki Organisasi

- Diagram hierarki pegawai berdasarkan hasil query:



III. Tugas 3: Analisis KPI Pegawai

1. **Staff Berprestasi Berdasarkan Jumlah Customer**
    - Query untuk mendapatkan jumlah customer tiap pegawai.

```
SELECT employee.employeeNumber AS id_staff,

       CONCAT(employee.firstName, ' ', employee.lastName) AS
staff,

       COUNT(cust.customerNumber) AS total_cust

FROM employees employee

LEFT JOIN customers cust ON employee.employeeNumber =
cust.salesRepEmployeeNumber

GROUP BY employee.employeeNumber

ORDER BY total_cust DESC

LIMIT 1;
```

id_staff	staff	total_cust
1401	Pamela Castillo	10

2. **Ranking Pegawai Berdasarkan KPI "Jumlah Customer"**
    - Penyusunan ranking pegawai dengan customer terbanyak.

```
WITH EmployeeHierarchy AS (

    SELECT

        manager.employeeNumber AS id_manager,

        CONCAT(manager.firstName, ' ', manager.lastName) AS
Manager,
```

```
        employee.employeeNumber AS id_staff,

        CONCAT(employee.firstName, ' ', employee.lastName) AS
staff,

        COUNT(cust.customerNumber) AS total_cust

    FROM employees employee

    JOIN employees manager ON employee.reportsTo =
manager.employeeNumber

    LEFT JOIN customers cust ON employee.employeeNumber =
cust.salesRepEmployeeNumber

    GROUP BY employee.employeeNumber

),

ManagerCustomer AS (

    SELECT

        e.id_manager,

        e.Manager,

        SUM(e.total_cust) AS total_customers

    FROM EmployeeHierarchy e

    GROUP BY e.id_manager, e.Manager

)

SELECT id_manager, Manager, total_customers

FROM ManagerCustomer

ORDER BY total_customers DESC;
```

id_manager	Manager	total_customers
1102	Gerard Bondur	46
1143	Anthony Bow	39
1088	William Patterson	10
1056	Mary Patterson	5
1002	Diane Murphy	0
1621	Mami Nishi	0

3. **Ranking Pegawai Berdasarkan KPI "Jumlah Omset"**
   - Query untuk menghitung total omset per pegawai.

```
SELECT employee.employeeNumber AS id_staff,

       CONCAT(employee.firstName, ' ', employee.lastName) AS
staff,

       SUM(payments.amount) AS total_omset

FROM employees employee

JOIN customers cust ON employee.employeeNumber =
cust.salesRepEmployeeNumber

JOIN payments ON cust.customerNumber = payments.customerNumber

GROUP BY employee.employeeNumber

ORDER BY total_omset DESC;
```

id_staff	staff	total_omset ▼ 1
1370	Gerard Hernandez	1112003.81
1165	Leslie Jennings	989906.55
1401	Pamela Castillo	750201.87
1501	Larry Bott	686653.25
1504	Barry Jones	637672.65
1323	George Vanauf	584406.80
1337	Loui Bondur	569485.75
1611	Andy Fixter	509385.82
1612	Peter Marsh	497907.16
1286	Foon Yue Tseng	488212.67
1621	Mami Nishi	457110.07
1216	Steve Patterson	449219.13
1702	Martin Gerard	387477.47
1188	Julie Firrelli	386663.20
1166	Leslie Thompson	347533.03

4. **Jumlah Field yang Dibutuhkan**

KPI	Jumlah Field
Jumlah customer yang bertransaksi	1 (total_cust)
Jumlah omset yang didapat	1 (total_omset)

Total **2 field** dibutuhkan dalam analisis ini.

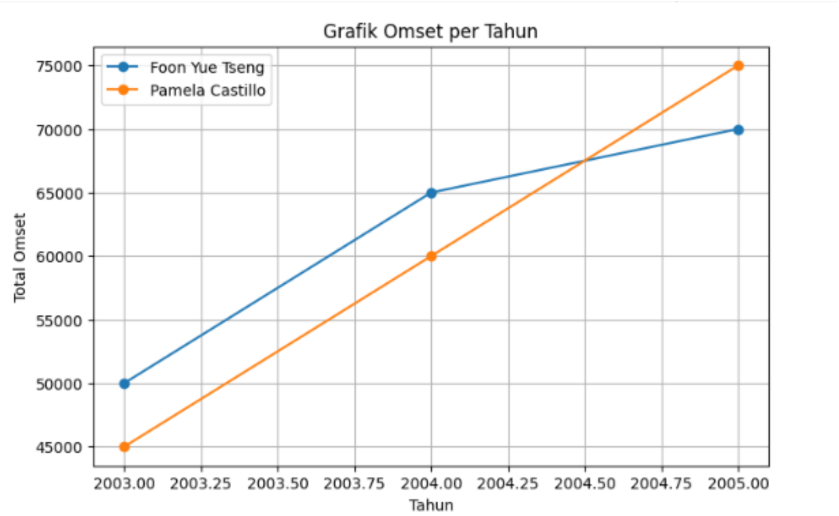
5. **Report Tahunan KPI "Jumlah Omset"**

a. **Data Omset Per Tahun untuk Pegawai Tertentu**

employeeNumber	staff	tahun 1	total_omset 2
1401	Pamela Castillo	2003	317104.78
1286	Foon Yue Tseng	2003	221887.03
1401	Pamela Castillo	2004	409910.07
1286	Foon Yue Tseng	2004	237255.26
1286	Foon Yue Tseng	2005	29070.38
1401	Pamela Castillo	2005	23187.02

b. **Grafik Omset Pegawai**

- Menampilkan grafik garis berdasarkan data di atas.



VII. Studi Kasus: Dashboard Penjualan per Cabang

1. **Analisis Field yang Diperlukan**
   - Data yang dibutuhkan untuk menampilkan omset per cabang.

Field	Keterangan
`branchName`	Nama cabang
`paymentDate`	Tahun transaksi
`amount`	Total omset dari transaksi
`customerNumber`	ID customer yang melakukan transaksi
`salesRepEmployeeNumber`	ID pegawai yang menangani customer
`employeeNumber`	ID pegawai (untuk melihat hierarki)

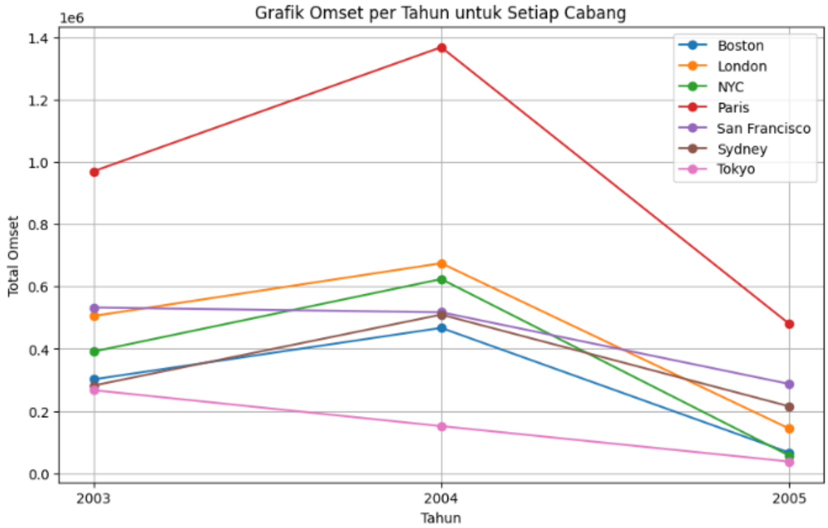
Query untuk Menampilkan Data Penjualan per Cabang

```
SELECT branchName, YEAR(saleDate) AS Tahun, SUM(saleAmount) AS
TotalOmset
FROM sales
GROUP BY branchName, YEAR(saleDate);
```

Hasil Report Cabang

Nama Cabang	2003	2004	2005
Boston	301,781.38	467,177.07	66,923.88
London	505,384.85	674,815.75	144,125.30
NYC	391,175.53	623,872.78	57,571.16
Paris	969,959.90	1,368,458.96	480,750.04
San Francisco	532,681.13	517,408.62	287,349.83
Sydney	281,985.51	509,833.62	215,473.85
Tokyo	267,249.40	151,761.45	38,099.22

4. **Grafik Penjualan Cabang**
    - Grafik berdasarkan data di atas.



VIII. Soal Bonus: Report Alternatif

1. **Analisis Data dan Query Alternatif**

    OLTP (Online Transaction Processing) menyimpan transaksi bisnis secara real-time. Dalam konteks **LegendVehicle**, beberapa tabel yang relevan untuk analisis adalah:

    - **customers** → Berisi informasi pelanggan
    - **employees** → Menyimpan data pegawai dan relasi dengan pelanggan
    - **offices** → Informasi cabang perusahaan
    - **payments** → Merekam pembayaran dari pelanggan
    - **orders & orderdetails** → Menyimpan informasi pesanan dan produk

    Query Alternatif

a) Menampilkan Jumlah Transaksi per Cabang per Tahun

```
SELECT

    offices.city AS Nama_Cabang,

    YEAR(orders.orderDate) AS Tahun,

    COUNT(orders.orderNumber) AS Total_Transaksi

FROM orders

JOIN customers ON orders.customerNumber = customers.customerNumber

JOIN employees ON customers.salesRepEmployeeNumber =
employees.employeeNumber
```

```
JOIN offices ON employees.officeCode = offices.officeCode

GROUP BY offices.city, Tahun

ORDER BY offices.city, Tahun;
```

Nama_Cabang	Tahun	Total_Transaksi
Boston	2003	9
Boston	2004	18
Boston	2005	5
London	2003	18
London	2004	24
London	2005	5
NYC	2003	14
NYC	2004	22
NYC	2005	3
Paris	2003	34
Paris	2004	49
Paris	2005	23
San Francisco	2003	17
San Francisco	2004	17
San Francisco	2005	14
Sydney	2003	12
Sydney	2004	15
Sydney	2005	11
Tokyo	2003	7
Tokyo	2004	6
Tokyo	2005	3

b) Menampilkan Total Omset per Cabang per Tahun

```
SELECT

    offices.city AS Nama_Cabang,

    YEAR(payments.paymentDate) AS Tahun,

    SUM(payments.amount) AS Total_Omset

FROM payments

JOIN customers ON payments.customerNumber = customers.customerNumber

JOIN employees ON customers.salesRepEmployeeNumber =
employees.employeeNumber
```

```
JOIN offices ON employees.officeCode = offices.officeCode

GROUP BY offices.city, Tahun

ORDER BY offices.city, Tahun;
```

Nama_Cabang	Tahun ▲ 2	Total_Omset
Boston	2003	301781.38
Boston	2004	467177.07
Boston	2005	66923.88
London	2003	505384.85
London	2004	674815.75
London	2005	144125.30
NYC	2003	391175.53
NYC	2004	623872.78
NYC	2005	57571.16
Paris	2003	969959.90
Paris	2004	1368458.96
Paris	2005	480750.04
San Francisco	2003	532681.13
San Francisco	2004	517408.62
San Francisco	2005	287349.83
Sydney	2003	281985.51
Sydney	2004	509833.62
Sydney	2005	215473.85
Tokyo	2003	267249.40
Tokyo	2004	151761.45
Tokyo	2005	38099.22

2. **Hasil Report**

Nama Cabang	**2003**	**2004**	**2005**
Boston	9	17	3
London	16	22	4
NYC	14	21	2
Paris	30	44	13
San Francisco	15	13	6
Sydney	11	13	6
Tokyo	5	6	3

Tabel ini menunjukkan jumlah total transaksi untuk setiap cabang dari tahun **2003 hingga 2005**.

3. **Grafik Report Alternatif**
   - Menampilkan grafik berdasarkan data OLTP.